UNIT - II

SHEAR FORCE & BENDING MOMENT

Mechanics of Solids (23A03302) - handwritten class notes

Topics covered in this unit

- Definition of a beam, types of beams, supports and loads
- Concept of shear force (S.F.) and bending moment (B.M.), sign conventions
- S.F. and B.M. diagrams: cantilever, simply supported and overhanging beams
- Point loads, u.d.l., uniformly varying load (u.v.l.) and combinations
- Point of contra-flexure
- Relation between S.F., B.M. and rate of loading

Tip: SFD and BMD questions carry good marks. Learn the standard results and the step-by-step method.

1. Beam - definition

A beam is a structural member, usually horizontal, whose length is large compared with its cross-section and which is loaded transverse (perpendicular) to its axis. Its main action is bending.

- Loads on beams cause internal shear force and bending moment at every section.
- Beam analysis: find reactions → find S.F. and B.M. → draw diagrams → design.

2. Types of beams

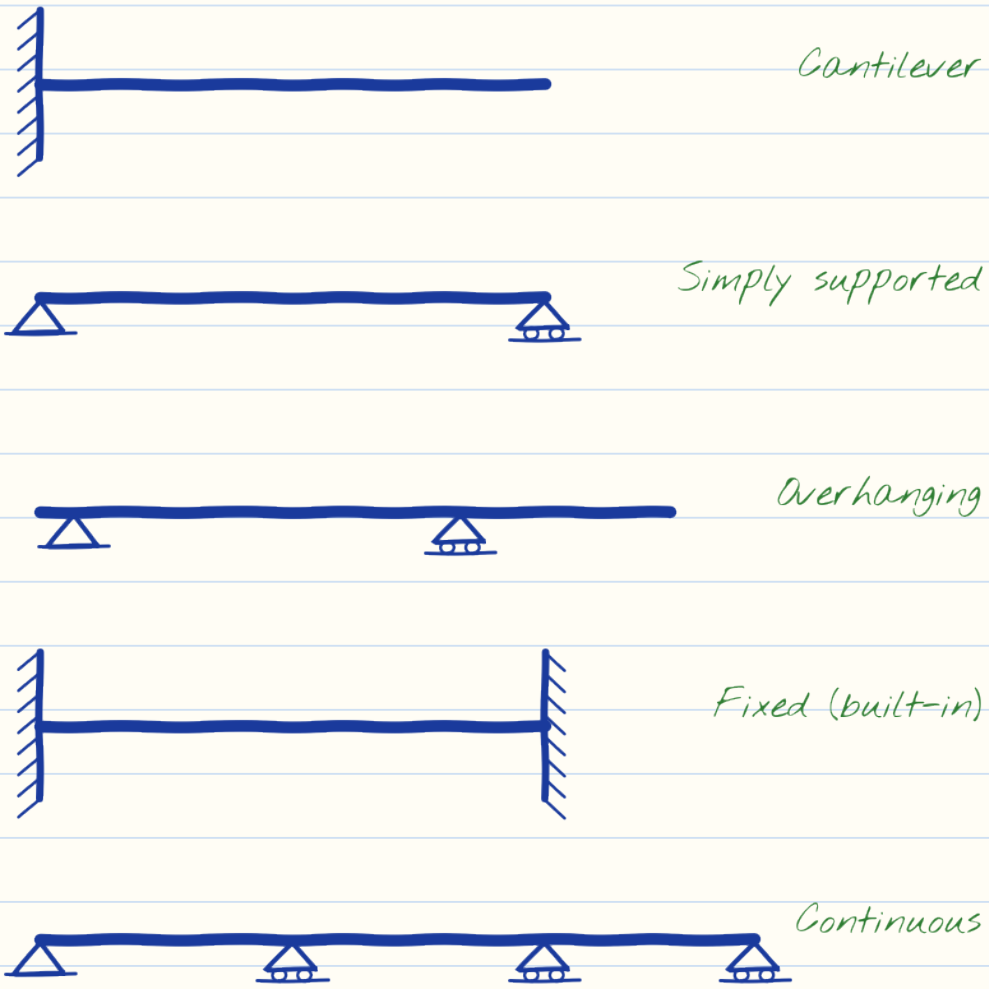


Fig: Common types of beams

- Cantilever: one end fixed, the other end free.
- Simply supported: rests on a hinge at one end and a roller at the other.
- Overhanging: a simply supported beam extended beyond one or both supports.
- Fixed (built-in): both ends fixed - restrained against rotation.
- Continuous: supported at more than two points.

Statically determinate beams: reactions found from equilibrium alone (cantilever, simply supported, overhanging). Fixed and continuous beams are statically indeterminate.

3. Types of supports

Support	Reactions	Restrains
Roller	one (vertical)	vertical movement
Hinge (pin)	two (V and H)	vertical + horizontal
Fixed	V, H and moment	movement + rotation

4. Types of loads

- Point (concentrated) load W - acts at a point, unit KN.
- Uniformly distributed load (u.d.l.) w - constant intensity, KN/m. Equivalent point load = w x length, acting at the centre of the load.
- Uniformly varying load (u.v.l.) - intensity varies linearly. Triangular load: total = ½ x base x peak, acts at 1/3 from the larger end.
- Couple (moment) applied at a point, unit KN·m.

5. Shear force and bending moment

Shear force at a section = the algebraic sum of all vertical forces acting on either side of the section.

Bending moment at a section = the algebraic sum of the moments of all forces on either side of the section about that section.

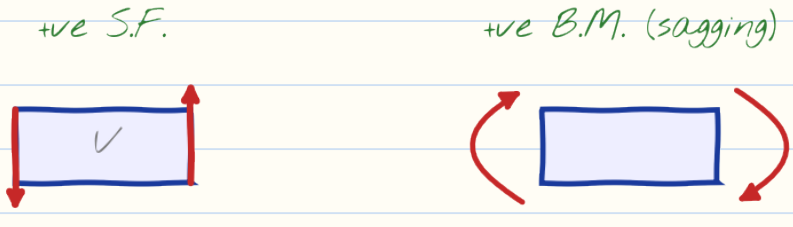


Fig: Sign conventions

- S.F. positive: left part tends to move up relative to right (clockwise shear). Forces on the left upward, or on the right downward, give +ve S.F.

- B.M. positive (sagging): bends the beam concave upward - tension at bottom. Negative = hogging, tension at the top.

Quick rule (working from the left): upward force → +ve S.F.; downward force → -ve S.F. Moments of upward forces to the left give +ve (sagging) B.M.

6. Relations between w, S.F. and B.M.

Consider a small element dx carrying load w per unit length:

$$\frac{dV}{dx} = -w \qquad \frac{dM}{dx} = V \qquad \frac{d^2M}{dx^2} = -w$$

- The slope of the S.F. diagram = (−) load intensity.
- The slope of the B.M. diagram = S.F. at that section.
- B.M. is maximum (or minimum) where S.F. = 0 or changes sign.
- Change in S.F. between two sections = area of the load diagram between them.
- Change in B.M. between two sections = area of the S.F. diagram between them.
- At a point load, S.F. jumps by the load; at an applied couple, B.M. jumps by the couple.

Load on span	S.F.D.	B.M.D.
No load	horizontal line	inclined straight line
Point load	sudden jump	change of slope (kink)
u.d.l.	inclined straight line	parabola
u.v.l.	parabola	cubic curve

7. Point of contra-flexure

The point of contra-flexure (point of inflexion) is the section where the B.M. changes sign - i.e. where B.M. = 0 and the curvature of the beam reverses (sagging to hogging).

- Found by equating the B.M. expression to zero.

- Exists mostly in overhanging and fixed/continuous beams. Cantilevers and simply supported beams with downward loads have none.

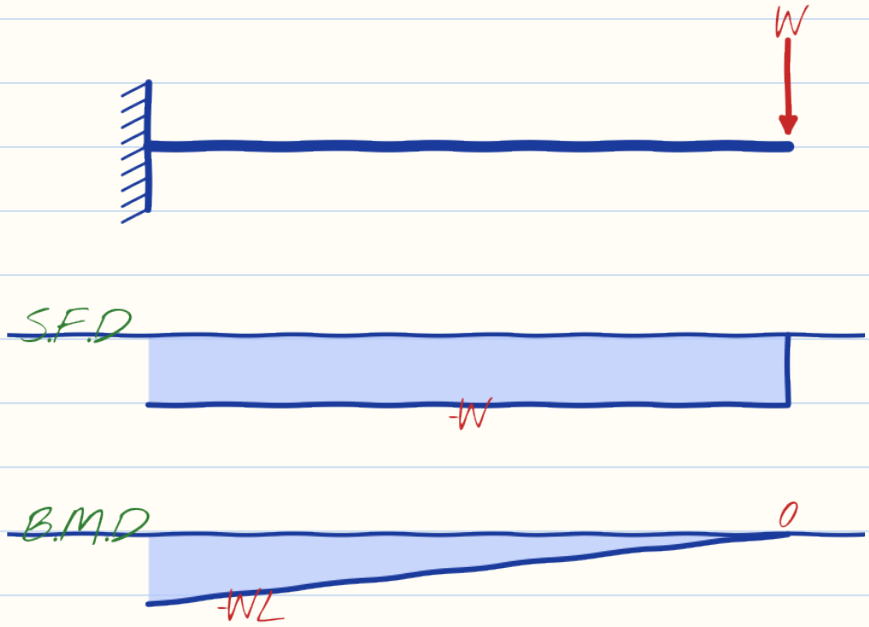
8. Procedure for drawing S.F.D. and B.M.D.

- 1. Replace distributed loads by equivalent point loads to find reactions ($\Sigma V = 0$, $\Sigma M = 0$).
- 2. Divide the beam into portions at every point load, support, and start/end of u.d.l.
- 3. Write V and M for each portion using one side of the section.
- 4. Calculate V and M at the ends of each portion and plot.
- 5. Join with straight line / curve as per the load. Put sign, value and units.
- 6. Locate the section of zero shear for the maximum bending moment.

9. Cantilever beams

(a) Point load W at the free end

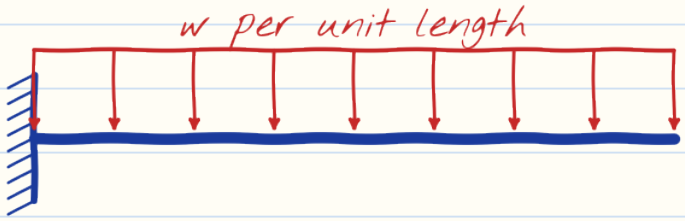
Take x from the free end: $V = -W$ (constant), $M = -Wx$.



$V_{max} = -W$ $M_{max} = -WL$ (at the fixed end)

(b) u.d.l. w over the whole span

$V = -wx$ (straight line), $M = -wx^2/2$ (parabola).







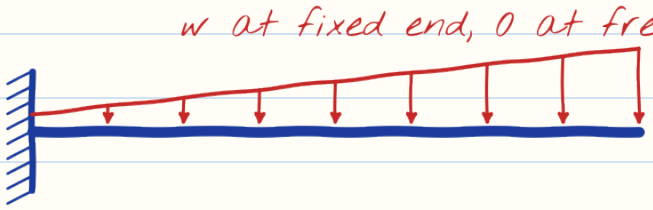
$V_{max} = -wL$ $M_{max} = -wL^2/2$

(c) u.v.l. – zero at the free end, w per metre at the fixed end

Intensity at x from free end = wx/L. Load up to $x = \frac{1}{2}(wx/L)x$.

$V = -wx^2/2L$ $M = -wx^3/6L$

at fixed end: $V = -wL/2$, $M = -wL^2/6$







Example 1: cantilever with combined loads

A cantilever of length 4 m carries a point load of 10 KN at the free end and a u.d.l. of 5 KN/m over the whole length. Find V and M at the fixed end.

V at free end = − 10 KN

V at fixed end = − (10 + 5×4) = − 30 KN

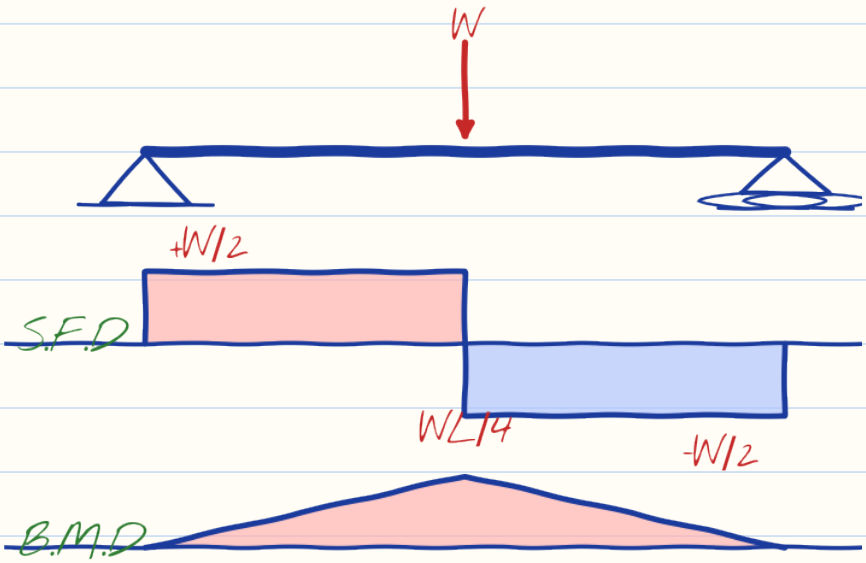
M at fixed end = − (10×4 + 5×4×2) = − (40 + 40) = − 80 KN·m

Both diagrams lie below the base line (hogging): V is straight (u.d.l.), M is a parabola.

10. Simply supported beams

(a) Central point load W

$R_A = R_B = W/2$. For $x < L/2$: $V = W/2$, $M = W x/2$. For $x > L/2$: $V = -W/2$.



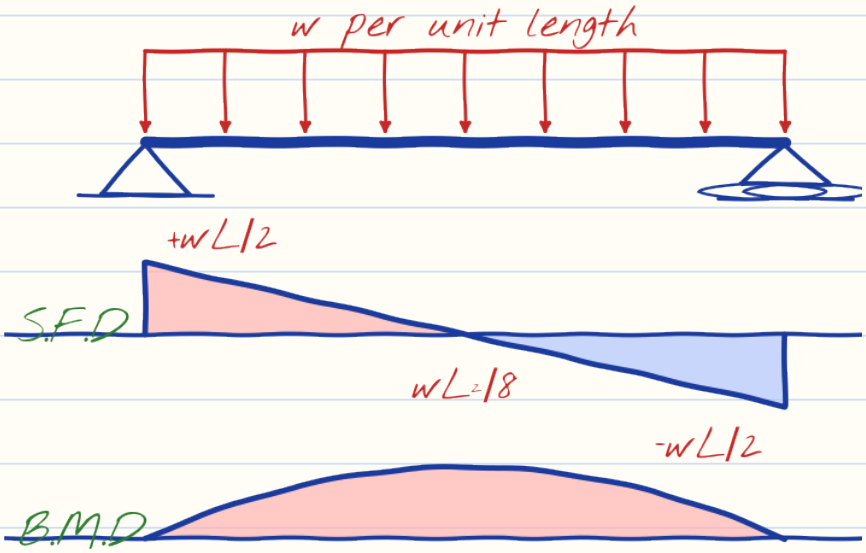
$$M_{max} = W L / 4 \text{ at the centre}$$

For an eccentric load W at distance a from A and b from B ($L = a + b$):

$$R_A = W b / L \qquad R_B = W a / L$$
$$M_{max} = W a b / L \text{ (under the load)}$$

(b) u.d.l. w over the full span

$R_A = R_B = wL/2$. $V = wL/2 - w\,x$ (zero at $x = L/2$). $M = wL\,x/2 - w\,x^2/2$.



$$M_{max} = w\,L^2 / 8 \text{ at the centre}$$

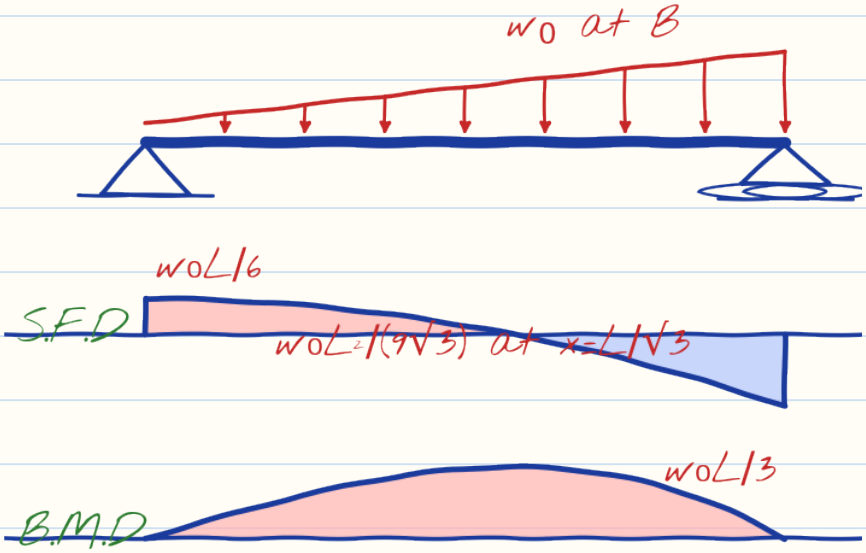
(c) u.v.l. – zero at A, w_0 per metre at B

Total load = $w_0\,L/2$ acting at $L/3$ from the heavy end B.

$$R_A = w_0L/6 \qquad R_B = w_0L/3$$

$V = w_0L/6 - w_0x^2/2L$ (zero at $x = L/\sqrt{3} = 0.577\,L$)

$M_{max} = w_0L^2/(9\sqrt{3}) = 0.0642\,w_0L^2$



Example 2: simply supported beam with point load and u.d.l.

Beam AB = 8 m. Point load 20 KN at 2 m from A; u.d.l. 5 KN/m from 4 m to 8 m (over 4 m). Draw SFD and BMD.

Moments about A: $R_B \times 8 = 20 \times 2 + (5 \times 4) \times 6 = 160 \Rightarrow R_B = 20$ KN

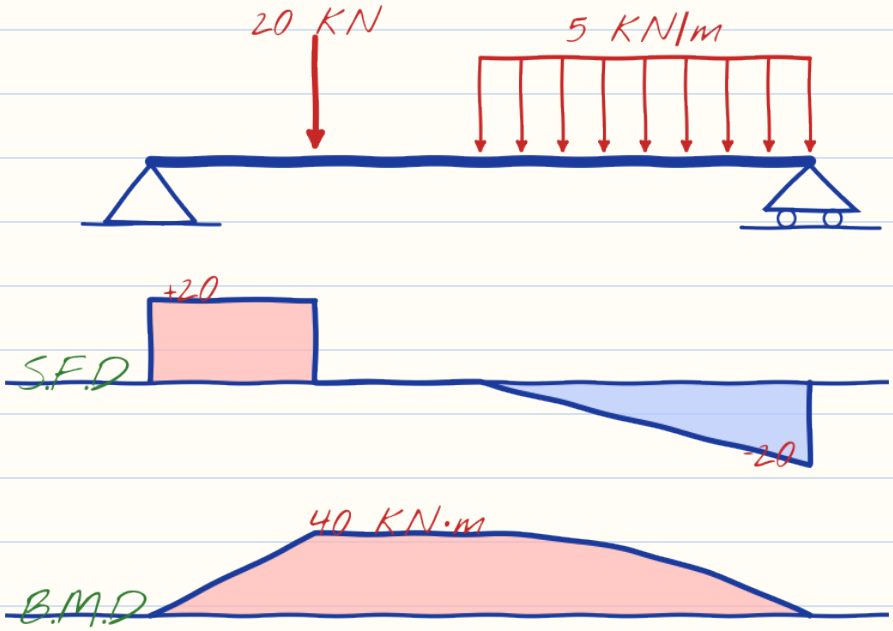
$R_A = 20 + 20 - 20 = 20$ KN

S.F.: A: +20; just left of C (2 m): +20; just right: 0; constant 0 up to 4 m;

then falls at 5 KN/m to $0 - 5 \times 4 = -20$ KN at B.

B.M.: at 2 m = $20 \times 2 = 40$; at 4 m = 40 (S.F. is zero between 2 and 4 m);

at 8 m = $40 - 5 \times 4^2/2 = 0$ ✓. $M_{max} = 40$ KN·m between C and D.



II. Overhanging beams

An overhanging beam has a part projecting beyond a support. The overhang gives a negative (hogging) B.M. over the support and hence a point of contra-flexure in the span.

Example 3: overhanging beam with u.d.l.

A beam 8 m long rests on supports A and B 6 m apart (overhang BC = 2 m) and carries a u.d.l. of 10 KN/m over the whole 8 m. Draw SFD and BMD, locate the point of contra-flexure.

Total load = $10 \times 8 = 80$ KN at 4 m from A.

$\Sigma M_A = 0$: $R_B \times 6 = 80 \times 4 \Rightarrow R_B = 53.33$ KN, $R_A = 26.67$ KN

S.F.: A = +26.67; just left of B = 26.67 − 60 = −33.33;

just right of B = −33.33 + 53.33 = +20; at C = 20 − 20 = 0 ✓

B.M. in span: $M = 26.67x - 5x^2$ (x from A)

V = 0 at $x = 26.67/10 = 2.667$ m: $M_{max} = 35.56$ KN·m

M at B = 26.67×6 − 5×36 = −20 KN·m (hogging) = − 10×2×1 ✓

Contra-flexure: $26.67x - 5x^2 = 0 \Rightarrow x = 5.33$ m from A

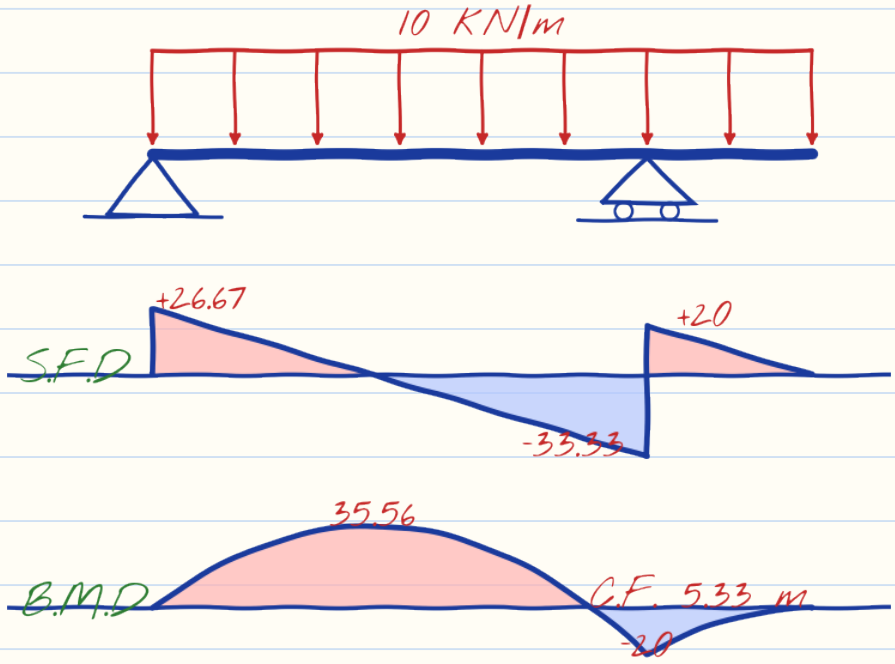


Fig. S.F.D. and B.M.D. of Example 3

12. Effect of an applied couple

- A couple M_0 does not change the S.F. at its point.

- The B.M. diagram jumps by M_0 at the point: a clockwise couple on the left part gives an upward jump (sagging +ve).
- Reactions of a simply supported beam with a couple M_0: $R_A = -M_0/L$, $R_B = +M_0/L$ (clockwise couple).

More solved problems

Example 4: S.S. beam, two point loads

Span AB = 6 m; 10 KN at 2 m and 20 KN at 4 m from A.

$R_B \times 6 = 10\times2 + 20\times4 = 100 \Rightarrow R_B = 16.67$ KN, $R_A = 13.33$ KN

S.F.: 0–2 m: +13.33; 2–4 m: 13.33 − 10 = +3.33; 4–6 m: 3.33 − 20 = −16.67

B.M.: at 2 m = 13.33×2 = 26.67; at 4 m = 26.67 + 3.33×2 = 33.33; at B = 33.33 − 16.67×2 = 0 ✓

$M_{max} = 33.33$ KN·m under the 20 KN load (S.F. changes sign there).

Example 5: cantilever with partial u.d.l.

Cantilever 3 m: u.d.l. 4 KN/m over the 2 m next to the fixed end, point load 6 KN at the free end.

Free end to 1 m: V = −6 (constant), then V decreases linearly to $-6 - 4\times2 = -14$ KN at the wall.

M at start of u.d.l. (1 m from free end) = $-6 \times 1 = -6$ KN·m

M at wall = $-(6\times3 + 8\times1) = -26$ KN·m (u.d.l. resultant 8 KN acts 1 m from the wall)

Example 6: S.S. beam with triangular load

Span 6 m, load increases from zero at A to 12 KN/m at B.

Total W = ½ × 6 × 12 = 36 KN, acting 2 m from B (4 m from A).

$R_A = 36 \times 2/6 = 12$ KN, $R_B = 24$ KN

$V = 12 - 12x^2/(2\times6) = 12 - x^2$; $V = 0$ at $x = \sqrt{12} = 3.464$ m

$M_{max} = 12 \times 3.464 - 3.464^3/3 = 41.57 - 13.86 = 27.71$ KN·m (= $w_0L^2/9\sqrt{3}$ ✓)

Tips and common mistakes

- Always check: reactions sum to the total load and $\Sigma M = 0$ about the other support.
- S.F. diagram must close to zero at the free end / right support. B.M. must be zero at simple supports and free ends.
- Distributed loads: convert to a point load only for finding reactions or for a section beyond the load.
- For a section inside a u.d.l., use only the part of the load to the left (or right) of the section.
- Area method: M at a point = M at the previous point + area of S.F. diagram between them.
- Do not forget the units - KN, KN/m, KN·m.

13. Shape of the diagrams - summary

Beam & load	V_{max}	M_{max}
Cantilever, W at end	W	W L
Cantilever, u.d.l.	w L	$w L^2 / 2$
Cantilever, u.v.l. (0 at free end)	$w L / 2$	$w L^2 / 6$
S.S., central W	$W / 2$	$W L / 4$
S.S., u.d.l.	$w L / 2$	$w L^2 / 8$
S.S., u.v.l. (0 at A)	$w L / 3$	$0.0642\, w L^2$

Practice problems (answers given)

- S.S. beam, span 10 m, u.d.l. 2 KN/m over the whole span. Find reactions and M_{max}. Ans: 10 KN each, 25 KN·m
- Cantilever 2 m with a 5 KN point load at the free end. Find M at the fixed end. Ans: -10 KN·m
- S.S. beam, span 4 m, central point load 8 KN. Find reactions and M_{max}. Ans: 4 KN each, 8 KN·m

- Cantilever 3 m with u.d.l. 6 KN/m. Find V and M at the fixed end. Ans: -18 KN, -27 KN·m
- S.S. beam, span 6 m, 24 KN at 2 m from A. Find R_A, R_B, M_{max}. Ans: 16 KN, 8 KN, 32 KN·m

Important questions

- Define S.F. and B.M. State the sign conventions.
- Derive the relation between load intensity, shear force and bending moment.
- Draw SFD and BMD for a simply supported beam with u.d.l. and show $M_{max} = wL^2/8$.
- What is a point of contra-flexure? Locate it for an overhanging beam.
- Draw the SFD and BMD for a cantilever loaded with a point load, u.d.l. and u.v.l.